PCB Design Report

Fixed-Bias Bistable Multivibrator

(Eccles-Jordan Flip-Flop) — Schematic Design, PCB Layout, Track Routing and Gerber Fabrication Data

Document type:	Electronic Design & PCB Layout Report
Board revision:	Rev A
Layers:	2 (Top / Bottom)
Board size:	55 mm x 42 mm
Fabrication data:	Gerber RS-274X + Excellon drill

1. Objective

To design the circuit of a Fixed-Bias Bistable Multivibrator (Eccles-Jordan flip-flop), calculate component values for reliable switching between its two stable states, lay out a two-layer printed circuit board for the circuit, route copper tracks for signal, supply and ground nets, and generate the Gerber/Excellon fabrication files needed to manufacture the board.

2. Circuit Theory

A bistable multivibrator has two stable states: with transistor Q1 ON (saturated) and Q2 OFF (cut-off), or vice-versa. It remains latched in whichever state it is in until an external trigger forces it to switch — unlike an astable multivibrator (free-running) or a monostable multivibrator (one stable state, one quasi-stable state).

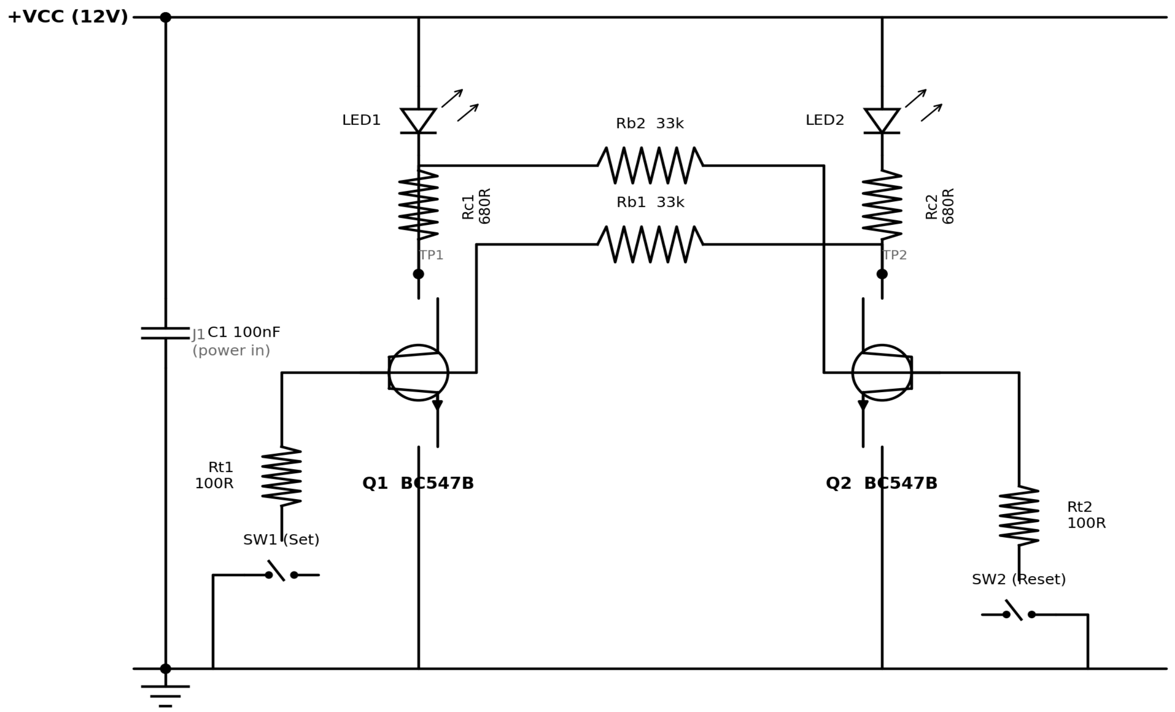
It is called **“fixed-bias”** because the two halves are cross-coupled purely through resistors (R_{B1}, R_{B2}) — the collector of each transistor drives the base of the other directly through a fixed resistive divider, with no capacitive (RC) coupling in the feedback path. This is what distinguishes it from the capacitor-coupled (astable) multivibrator. Because there is no capacitor to charge/discharge in the feedback loop, the circuit does not free-run — it needs a trigger (push-button, pulse, or steering diode network) to toggle.

Operating principle:

- Power-on: due to unavoidable small asymmetries, one transistor conducts slightly more than the other. Regenerative positive feedback through R_{B1}/R_{B2} rapidly drives that transistor into saturation and the other into cut-off.
- Say Q1 is ON (saturated): $V_{CE1(sat)} \approx 0.2$ V. This low voltage is fed through R_{B2} to the base of Q2, holding V_{BE2} below 0.6 V, so Q2 stays cut off.
- Q2 being OFF means its collector sits near V_{CC} (no drop across R_{C2}). This high voltage is fed through R_{B1} to the base of Q1, supplying base current well in excess of what is needed for saturation, so Q1 stays firmly ON.
- The state is latched by this cross-feedback. Pressing SW1 momentarily pulls Q1's base to ground, cutting Q1 off; the regenerative action then flips the circuit to the opposite state, where it again latches. SW2 flips it back.

3. Schematic Diagram

Fixed-Bias Bistable Multivibrator (Eccles-Jordan Flip-Flop) - Schematic



LED1/LED2 indicate which half is conducting (state indicator only — they are not required for circuit operation). TP1/TP2 are test points at the collectors for probing the output waveform on an oscilloscope. C1 is a supply decoupling capacitor.

4. Design Calculations

4.1 Assumptions

Supply voltage, V_{CC}	12 V
Transistor	BC547B (NPN), h_{FE(min)} = 110
V_{BE(sat)}	0.7 V
V_{CE(sat)}	0.2 V
LED forward voltage, V_F	2.0 V (in series with each collector resistor)
Desired overdrive factor, S = I_B/I_{B(min)}	≥ 2 (guarantees hard saturation)

4.2 Collector resistor (R_C)

Choosing a saturated collector current of about 14 mA (bright LED indication, low power draw):

$R_C = (V_{CC} - V_F - V_{CE(sat)}) / I_{C(sat)} = (12 - 2 - 0.2)$ / 14 mA ≈ 700 Ω → nearest E12 value: **680 Ω**

$I_{C(sat)} = (12 - 2 - 0.2)$ / 680 Ω = 14.4 mA

4.3 Cross-coupling base resistor (R_B)

Minimum base current required for saturation at worst-case h_{FE}:

$I_{B(min)} = I_{C(sat)} / h_{FE(min)}$ = 14.4 mA / 110 ≈ 131 µA

The base of the ON transistor is fed from V_{CC} (through the OFF transistor's un-loaded collector) via R_B:

$R_B \leq (V_{CC} - V_{BE(sat)}) / (S \times I_{B(min)}) = (12 - 0.7) / (2 \times 131$ µA) ≈ 43 kΩ → nearest E12 value: **33 kΩ**

Actual base drive: $I_B = (12 - 0.7)/33$ kΩ = 342 µA, giving an overdrive factor S = 342/131 ≈ 2.6 — comfortably in saturation across the h_{FE} spread of the batch.

Cut-off check: the OFF transistor's base is fed (through R_B) from the ON transistor's collector, which sits at $V_{CE(sat)} \approx 0.2$ V. Since 0.2 V < 0.6 V (V_{BE} turn-on threshold), no base current flows and the transistor stays cut off — the fixed resistive bias holds each half firmly in its assigned state.

4.4 Trigger network (R_t, SW)

R_{t1}/R_{t2} = 100 Ω limits the current spike when the push-button momentarily shorts the ON transistor's base to ground (protects the button contacts and the driving transistor). Pressing SW1 forces Q1 off and the circuit regeneratively latches with Q2 ON; SW2 does the reverse.

4.5 Decoupling

C1 = 100 nF ceramic capacitor placed across V_{CC}/GND close to the header, to suppress supply-rail transients generated during the fast regenerative switching action.

5. Bill of Materials (BOM)

Ref. Des.	Description	Value	Package	Qty
Q1, Q2	NPN transistor	BC547B	TO-92	2
Rc1, Rc2	Collector resistor	680 Ω, 1/4 W	Axial (THT)	2
Rb1, Rb2	Cross-coupling base resistor	33 kΩ, 1/4 W	Axial (THT)	2

Rt1, Rt2	Trigger current-limit resistor	100 Ω, 1/4 W	Axial (THT)	2
LED1, LED2	State-indicator LED	Red / Green, 5 mm	THT	2
SW1, SW2	Momentary push-button (Set/Reset)	6x6 mm tactile	THT	2
C1	Decoupling capacitor	100 nF ceramic	THT, 2.5 mm pitch	1
J1	Power input header	2-pin, 2.54 mm	THT header	1

6. PCB Design Specification

Parameter	Specification
Board outline	55 mm x 42 mm, rectangular, 4x M2.2 mounting holes (r = 1.1 mm) at corners
Layer stack	2-layer: Top copper (F.Cu) / FR-4 core, 1.6 mm / Bottom copper (B.Cu)
Copper weight	1 oz/ft^2 (35 µm) both layers
Component mounting	Through-hole (THT), single-sided placement on top silkscreen
Top layer (F.Cu)	All signal routing: VCC distribution, cross-coupling nets, trigger nets
Bottom layer (B.Cu)	Solid ground (GND) pour / plane, stitched to every GND pin with a via
Min. track width / spacing	0.3 mm / 0.3 mm (12 mil / 12 mil) — standard hobby-fab capable
Min. drill / annular ring	0.8 mm (THT component holes), 0.3 mm min. via drill, 0.15 mm annular ring
Soldermask	Standard green, both sides
Silkscreen	White, top side only — reference designators + polarity marks
Surface finish	HASL (lead-free) or ENIG

6.1 Track width calculation (IPC-2221)

Maximum current in this circuit is the LED/collector branch current, $I_{C(sat)} \approx 14.4$ mA — well under 1 A, so trace width is not current-limited here; the minimum width is instead set by the fabrication house's process capability. Using the standard external-layer IPC-2221 formula for a 10 °C temperature rise, 1 oz copper:

A trace of 0.3 mm (12 mil) width, 35 µm thick, can carry approximately 0.9–1.0 A for a 10 °C rise — over 60x this circuit's actual current. Track widths used in this layout:

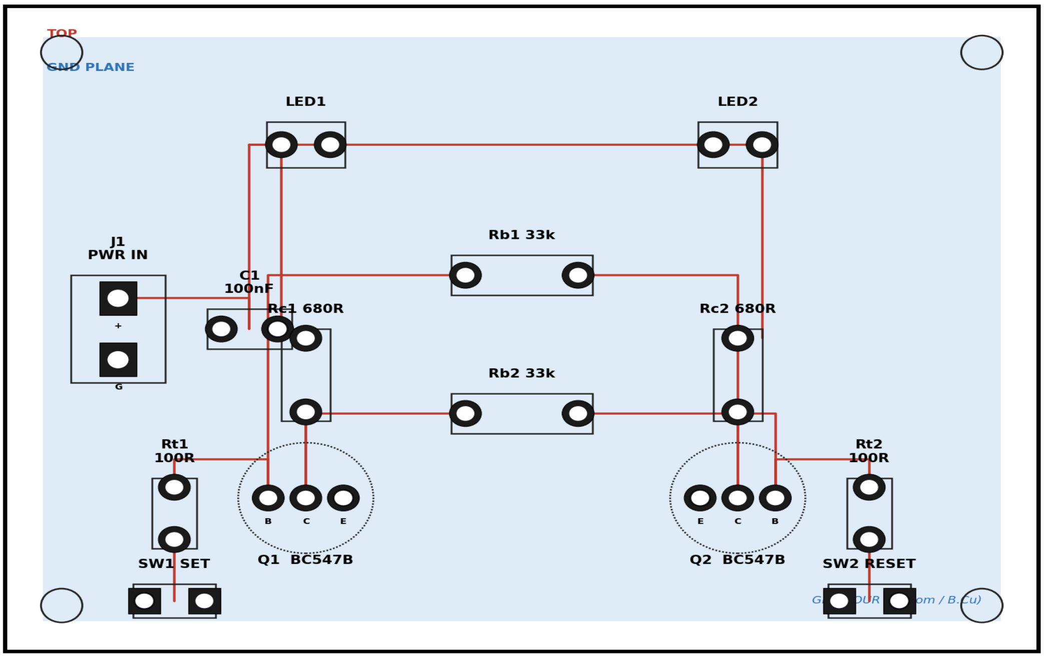
Net class	Track width	Rationale
VCC / GND	0.5 mm (20 mil)	Extra margin for supply robustness
Signal (base/collector cross-coupling, trigger)	0.3 mm (12 mil)	Low current, minimum fab-safe width

6.2 Layout strategy

- Symmetric, mirrored placement of the Q1 half and Q2 half about the board's vertical centre-line, reflecting the circuit's own symmetry and keeping the two cross-coupling tracks (R_{B1}, R_{B2}) short and equal in length.
- LED1/LED2 placed at the top edge of the board for visibility when the board is mounted in an enclosure; SW1/SW2 placed at the bottom edge for easy access.
- A solid ground pour is used on the bottom layer instead of routed GND traces, minimising ground impedance and providing shielding for the regenerative switching edges.
- The power input header (J1) and decoupling capacitor (C1) are placed adjacent to each other, close to the incoming supply edge, to minimise supply-loop inductance.

7. PCB Layout and Routing

PCB Layout - Top Copper + Bottom GND Pour
Fixed-Bias Bistable Multivibrator | Board: 55 x 42 mm, 2-layer FR4



Top copper (F.Cu) - signal / VCC
Bottom copper pour (B.Cu) - GND
THT pad / via
Silkscreen outline (F.SilkS)

Red traces: top-layer (F.Cu) copper carrying VCC and all signal nets, Manhattan-routed at 90°/45° bends. Light-blue region: bottom-layer (B.Cu) solid ground pour, stitched to every ground pin through a via. Dotted circles: BC547B TO-92 outline (silkscreen).

8. Gerber Fabrication Data

Once track routing on the layout above is finalised inside a PCB CAD tool (e.g. KiCad, Eagle, Altium), the following industry-standard Gerber RS-274X layers and an Excellon drill file are exported and sent to the fabrication house as a single ZIP archive:

File (KiCad convention)	Layer	Content
FixedBiasBistable-F_Cu.gtl	Top copper	Component-side signal + VCC tracks, pads
FixedBiasBistable-B_Cu.gbl	Bottom copper	Ground plane pour
FixedBiasBistable-F_SilkS.gto	Top silkscreen	Reference designators, polarity marks, outlines
FixedBiasBistable-B_SilkS.gbo	Bottom silkscreen	(unused / board title)
FixedBiasBistable-F_Mask.gts	Top soldermask	Soldermask openings over top pads
FixedBiasBistable-B_Mask.gbs	Bottom soldermask	Soldermask openings over bottom pads/vias
FixedBiasBistable-Edge_Cuts.gko	Board outline	Mechanical board edge + mounting holes
FixedBiasBistable.drl	Drill (Excellon)	All THT component holes + via drills

8.1 Recommended workflow to produce the manufacturable files

This report was generated in a text/graphics-only environment that does not have a PCB CAD package (KiCad/Eagle/Altium) installed, so the Gerber ZIP itself could not be plotted here. The schematic and the routed-layout diagram above are drawn to the exact netlist, component values, footprints and track topology needed — recreating them in a CAD tool is a direct, mechanical step. The free, open-source workflow is:

1. Install KiCad (kicad.org) 7.x or later.
2. In the Schematic Editor, place Q1, Q2 (BC547), Rc1/Rc2, Rb1/Rb2, Rt1/Rt2, LED1/LED2, SW1/SW2, C1 and J1 using the values in Section 5, and wire them exactly as shown in Section 3.
3. Run ERC (Electrical Rules Check), then assign PCB footprints: TO-92 for Q1/Q2, THT resistor (400 mil pitch) for R's, 5 mm LED, 6x6 mm tactile switch, 2.54 mm THT for C1/J1.
4. Generate the netlist and switch to the PCB Editor (Update PCB from Schematic).
5. Set the board outline to 55 x 42 mm, place footprints per Section 7's layout, then route the top-layer tracks (0.3 mm signal / 0.5 mm power) as shown, and pour a bottom-layer GND zone.
6. Run DRC (Design Rules Check) with 0.3 mm clearance until zero errors remain.
7. File → Fabrication Outputs → Gerbers (select all copper, mask, silkscreen and Edge.Cuts layers, RS-274X, Protel filename extensions) and Fabrication Outputs → Drill Files (Excellon, PTH+NPTH merged as required by the fab house).
8. Zip all output files and upload to the PCB manufacturer (e.g. JLCPCB, PCBWay, OSH Park).

9. Testing and Verification

- Power the board at 12 V; one of LED1/LED2 should light immediately (state chosen at random by component tolerances) while the other stays off.
- Press SW1 — the lit LED should switch to the other side and stay there (latched) after the button is released.
- Press SW2 — the LED state should flip back.

- Probe TP1 and TP2 with an oscilloscope: each should show a clean two-level square-edge waveform (≈ 0.2 V when its transistor is ON/saturated, ≈ 9.6 V when OFF), confirming correct bistable operation with no oscillation between triggers.

End of report — Fixed-Bias Bistable Multivibrator PCB Design, Rev A.